

Scalability & Future Plan

Date	29/09/2026
Team ID	UNM1425EduGenie
Project Name	EduGenie
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current EduGenie solution can be scaled to support larger user bases, increased data requirements, and additional learning features in the future.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	No persistent database for storing user profiles and learning progress	Limits long-term progress tracking and personalized history	High
2	Dependency on Google Gemini API and internet connectivity	AI features may be affected by API availability, limits, or network issues	High
3	No user authentication and personalized dashboard	Users cannot maintain individual accounts or track their learning history	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Designed as a local web application using FastAPI and Uvicorn	Deploy on scalable cloud infrastructure and use multiple application instances for larger user loads
2	Data Storage	No persistent database is currently used	Integrate SQLite initially and PostgreSQL/MySQL for larger-scale deployments
3	Performance	Requests are processed through FastAPI and the Gemini API	Optimize API requests, introduce caching where appropriate, and use scalable server resources
4	Security	API key is stored in environment variables and not exposed to the frontend	Add user authentication, authorization, stronger input validation, and secure deployment practices

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	User login, profile management, and progress tracking	Short Term	Enables personalized user accounts and learning history
Phase 2	Database integration and personalized dashboard	Medium Term	Supports persistent learning data and progress visualization
Phase 3	Voice assistant and multilingual learning support	Medium Term	Improves accessibility and supports learners in multiple languages
Phase 4	Adaptive learning and continuous AI tutor capabilities	Long Term	Provides more personalized and adaptive learning experiences